

REVISION 2021

Engineering Workshop Practice

Comprehensive Handbook for First Year Diploma

Code: 2009

Credits: 1.5

Periods: 90 Hours

Type: Workshop

Course Outcomes (CO)

- **CO1:** Identify the safety precautions, tools and devices required to make carpentry joints. (18 Hours)
- **CO2:** Make use of various tools, machines, instruments and power tools used in the Fitting shop to make fitting joints. (18 Hours)
- **CO3:** Make use of various tools, machines, instruments and power tools used in the Welding shop to make welding joint. (18 Hours)
- **CO4:** Utilize different sheet metal tools and measuring instruments to make sheet metal joints. (18 Hours)

- **CO5:** Make use of various tools and accessories to practice electrical wiring, motor connection and soldering. (18 Hours)

Curriculum Map

Module	Shop / Topics	Hours	CO Mapping	Cognitive Level
1	Carpentry Shop	18	CO1	Applying
2	Fitting Shop	18	CO2	Applying
3	Welding Shop	18	CO3	Applying
4	Sheet Metal Shop	18	CO4	Applying
5	Electrical & Soldering	18	CO5	Applying

Module 1: Carpentry Shop

Carpentry involves the craft of cutting, shaping, and installing building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc.

M1.01: Safety & Tools

Personal safety, machine safety, and job safety. Identification of marking, measuring, and cutting tools.

M1.03: Open Halved Joint

Construction of a simple timber joint using marking, planning, and chiseling techniques.

SVG Illustration: Carpentry Tools

Wood Working Joint



Carpentry is the craft of working with wood to create structures, furniture, and decorative elements. It involves precise measurements, cutting, and joining of wood pieces. Carpentry is a traditional skill that has been passed down through generations. It is a versatile craft that can be used in many different ways, from building houses to creating art.

Module 2: Fitting Shop

Fitting is the process of assembling various parts to form a machine or a structure. It involves precise filing and finishing.

Fitting Operations

Filing, Hack sawing, Marking, and Finishing. Construction of V-joint or Square joint.

Fitting is a process of joining two or more pieces of metal or wood together to form a single unit. It involves the use of various tools and techniques to create a strong and durable joint. Fitting is a common operation in many industries, including construction, manufacturing, and repair work.

Module 3: Welding Shop

Welding is a fabrication process that joins materials, usually metals or thermoplastics, by using high heat to melt the parts together.

Arc Welding

Practicing Lap joint and Butt joint using Arc welding equipment.

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Module 5: Electrical & Soldering

Basic electrical wiring for domestic applications and electronic component soldering.

Electrical Wiring

One lamp controlled by one switch, Staircase wiring, and Series-Parallel connections.

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Practice Model Question Paper

Part A: Carpentry

Construct a Cross-Halved joint from the given timber specimen (Size: 150x50x25 mm).

Part B: Wiring

Draw the circuit diagram and wire a staircase connection using two-way switches.

Source Declaration

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